minimal electronics. This feature required the driver to do everything. To improve it, L1 autonomy was introduced with adaptive cruise control (ACC) and lane-keeping assist (LKA).

Currently, L4 autonomy is the trending norm, which requires minimum human intervention. It has a restriction, though, in the form of autonomous activation only in certain geofenced areas. In the future, L5 autonomy will be prevalent whose attractive feature will be the ability to operate under any condition with the driver simply being optional.

These multiple stages of autonomy indicate that the level of transportation has gradually increased, allowing enhancements in security standards. With the imminent arrival of autonomous vehicles, the new standard needs to be even more robust to face any associated technical and regulatory challenges.

Technical challenges

Complexity management. Automotive companies manufacture vehicles in compliance with the functional safety standard known as ISO26262. Although autonomy has significantly increased the safety level and ensured effective detection of any sensor failures for surrounding object localisation and categorisation, it remains a challenge with respect to minimising perception errors. Safety also plays an important role in ensuring that vehicle controls (engine, brakes, and steering systems) send out correct and timely actuator commands to the vehicle.

Technological limitations. Functional safety only focuses on detecting system failures and reacting to them. But what about an unintended scenario caused not due to system failures but because of technological limitations in sensor performance or inadequate training datasets to address different operating conditions?

Cybersecurity. Due to extensive vehicle-to-vehicle (V2V) communication, vehicle-to-pedestrian (V2P) communication, or vehicle-to-infrastructure (V2I) communication, fleets and customers must be protected from cybersecurity attacks. Connectivity additions here include IT backend systems and control interfaces between connected vehicles and external information sources. People with malicious intentions have a considerable interest in attacking these platforms.

Verification and validation. It refers to measuring safety for a secure system and how far the validity can be given for it.

Pillars of autonomous vehicles

- Sensors such as camera, radar, or lidar gather 360-degree information
- Neural network based autonomy software stack processes the data
- Output is fed into vehicle controls (engine, brakes, and steering systems)
- Electrical systems transmit signals across vehicle controls connecting to computer

Each pillar has its complexity. When integrated into a system, the level of safety significantly increases.



Fig. 1: Chain of implications involved in functional safety

Regulatory challenges

Proving customer safety. Before the on-road rollout of autonomous vehicles (AVs), customer safety should be subjected to rigorous testing. AVs are expected to reduce the crash rates, especially fatal ones, as compared to human-driven cars because machines operating them have less scope of errors. If AVs cannot justify this, there will be a significant setback to the adoption and rollout of AVs.

Liability. Current liability laws mandate manufacturers to be accountable for safety. Due to a steady shift in a vehicle's operation from manual

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